Alia Valentine

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Education

Cornell University — Ithaca, NY
PhD Applied Mathematics
Doctoral Advisor Anastasia Bizyaeva — August 2024 - Current

Michigan State University — East Lansing, MI
BS Physics and Mathematics, Graduated With Honor
Minor in Computational Mathematics, Science, and Engineering — September 2020 - April 2024

Selected Honors and Awards

Student travel award, IEEE CDC 2025 Conference	2025
Robert F. Smith Fellowship, Cornell University College of Engineering	2024
Thomas Osgood Award for Excellence in Academics and Research, MSU Physics	2024
Undergraduate Learning Assistant Award, Upper-Level Course, MSU Physics	2024
McCartney Scholarship Award, MSU Mathematics	2023

Publications

Preprints and Under Review

[P1] A. Valentine and A. Bizyaeva, *Effect of car-centric social norms on traffic congestion in a compartmental model of transit mode utilization*, arXiv:2611.13265 [math.DS], Nov. 2026.

Peer-reviewed conference papers

[C2] A. Valentine, D. M. Gordon, and A. Bizyaeva, "Mixed-feedback oscillations in the foraging dynamics of arboreal turtle ants", in *2025 IEEE 64th Conference on Decision and Control (CDC)*, ISSN: 2576-2370, Dec. 2025, pp. 3806–3811.

Presentations

Conference on Complex Systems 2026, Binghampton, USA	October 2026
OURFA^{2}M^2 2026 (panelist)	February 2026
IEEE Conference on Decision and Control 2025, Rio de Janeiro, Brazil	December 2025
Foundations of Adaptive Networks Working Group, Santa Fe Institute	May 2025

Selected Service and Outreach

IEEE NextCom Outreach Co-Chair	March 2026-Current
Cornell Mathematical Contest in Modeling Judge	Fall 2025, Fall 2026
Center for Applied Mathematics coffee hour organizer	September 2025 - Current
Science Theater demonstration maintenance lead	April 2023-May 2024

Teaching

Teaching Assistant, Cornell

Math Explorations (MATH 1300)	Spring 2026
Differential Equations for Engineers (MATH 2930)	Fall 2025

Undergraduate Learning Assistant, Michigan State University

Machine Learning Short Course	Summer 2023, Summer 2024
Electricity and Magnetism I (PHY 481)	Spring 2024
Classical Mechanics (PHY 321)	Spring 2024
Mathematical Modeling in Physics (PHY 415)	Fall 2023
Physics for Scientists and Engineers II (PHY 184)	Spring 2023, Spring 2022
Curriculum development for Physics for Scientists and Engineers II (PHY 184)	Fall 2022
Curriculum development for studio physics courses (PHY 173/PHY 174)	Summer 2022
Physics for Scientists and Engineers I	Fall 2021, Spring 2021